

10.1 How our food is produced

The production of food is an important part of ensuring a healthy population in countries around the world. This is achieved through agriculture, where people farm crops or livestock to produce food. There are different types and methods of farming. These fit into two main categories, described below.

What is being farmed

- **Arable farming:** farming crops like wheat, soya, or rice. The crops are grown in fields that farmers have usually ploughed for planting. Farmers generally have to replant and harvest the crops each year.
- **Pastoral farming:** farming livestock like cattle, sheep, or chickens.
- **Mixed farming:** farming both crops and livestock.

How it is being farmed

- **Subsistence farming:** growing crops and raising livestock to meet the needs of the farmer and their family on small plots of land. The farmer may sell small amounts of surplus (left over food) at local markets, but their main goal is to produce enough food to survive. Most subsistence farming takes place in LICs. This type of farming typically requires little financial investment, relies on unskilled labour, and has low yields.
- **Commercial farming:** producing crops and livestock on a large scale to sell for profit. It involves the use of mechanisation, chemical fertilisers, and pesticides to maximise **crop yields**. Commercial farmers will often specialise in one type of crop or animal and use advanced irrigation and storage systems. It is an important contributor to global food supplies.



Figure 10.1: A 76-year-old man works in his organic vegetable garden.



Figure 10.2: Large-scale maize farming.



Figure 10.3: Farming with sheep in a large open area where the animals graze in the field each day.



Figure 10.4: Farming in greenhouse tunnels where the climate and water is carefully controlled.



Figure 10.5: A woman works in the field growing food for her family.

Source **A**: Different types of farming

TASK 1: Study Source

- Which of the images represents subsistence farming? Explain your selection.
- Which of the images do you think represents the most expensive type of farming to set up? Explain your selection.
- What type of farming is represented in Figure 10.3? How does it differ from the farming in Figure 10.1, 10.2 and 10.4?

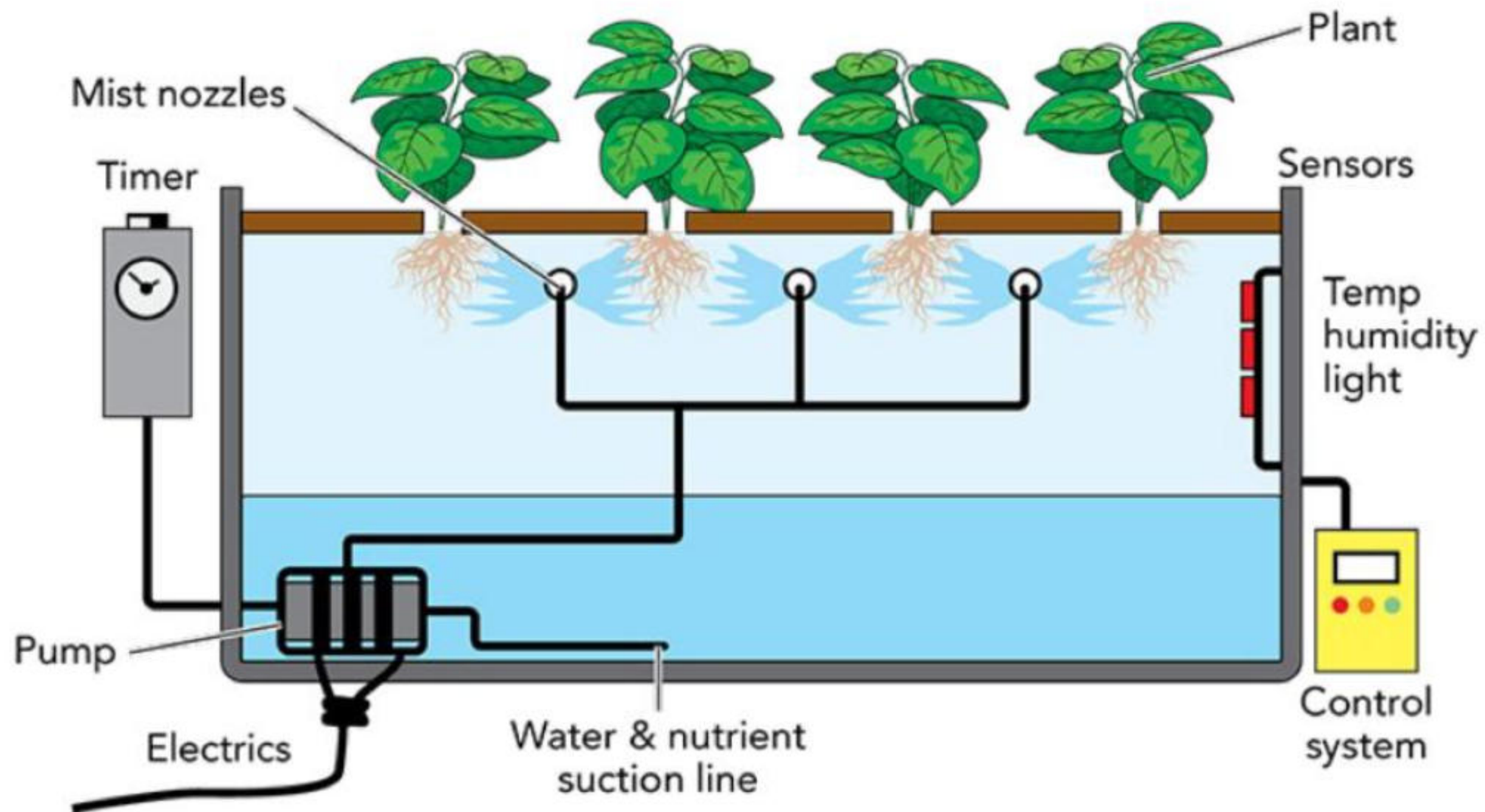
Aeroponics, hydroponics, and aquaponics

Population growth has resulted in an increase in the demand for food. At the same time, available farming land has decreased due to urbanisation, soil exhaustion from poor farming practices and climate change. In order to combat these challenges, farmers, and scientists have developed different farming technologies. These techniques include **aeroponics**, **hydroponics**, and **aquaponics**. These are all farming practices that are designed to produce as much food as possible, in as small a space as possible, while carefully managing the use of water.

Aeroponics

Aeroponics is the process of growing plants with the plants' root systems suspended in air. This takes place in a closed, climate-controlled environment. The plants are held in large grow racks, under artificial light. The farmers add liquid nutrients, such as nitrogen and potassium, to a large water store. This water is then delivered as a mist to the plant's roots (see Source B). As the nutrient-rich mist is delivered directly to the roots, the plants do not need to use up energy seeking the nutrients.

Although aeroponics was first used in the 1920s, it is only since the 1990s that it has developed as an option for growing food without soil. This is particularly important in environments where the availability of good soils is limited.



Source B: An aeroponics system

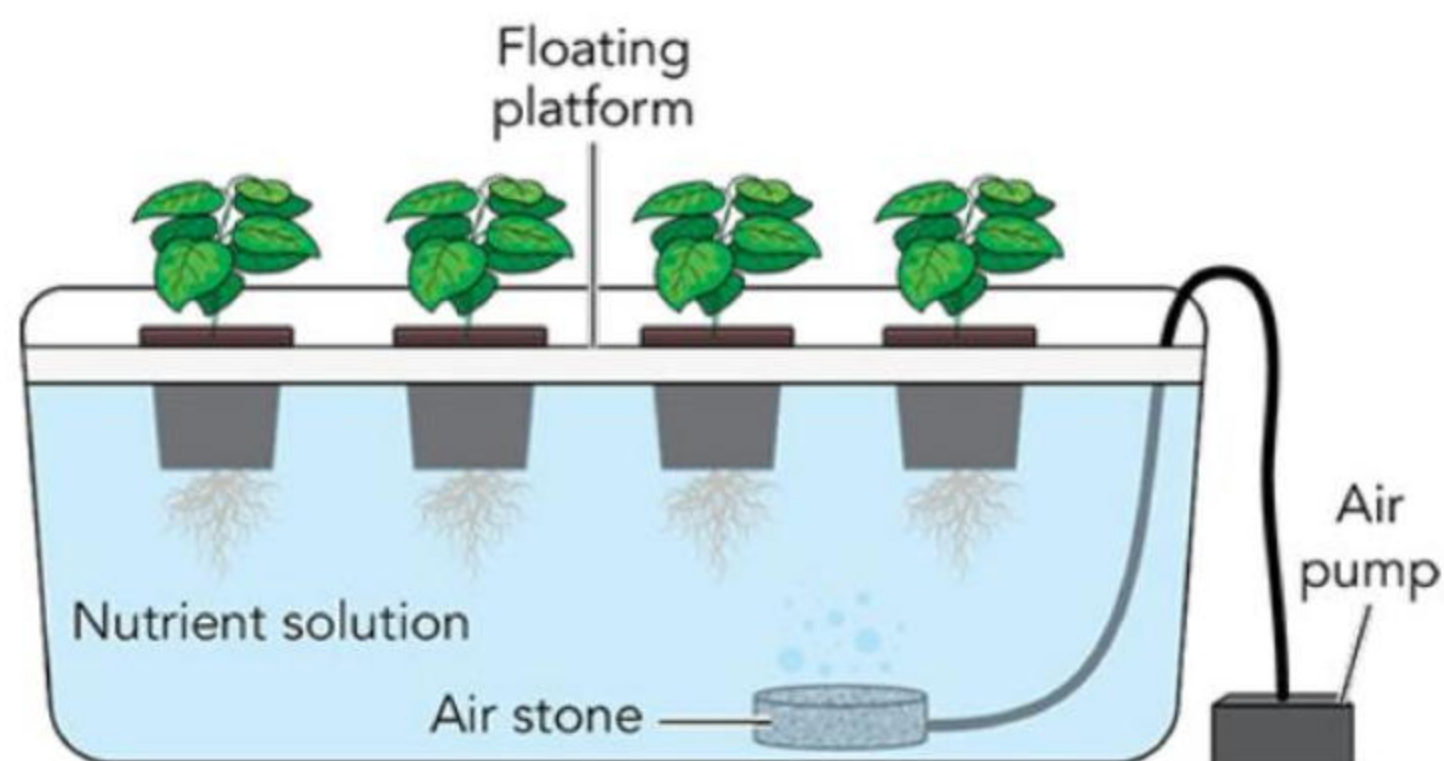
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Hydroponics and aquaponics

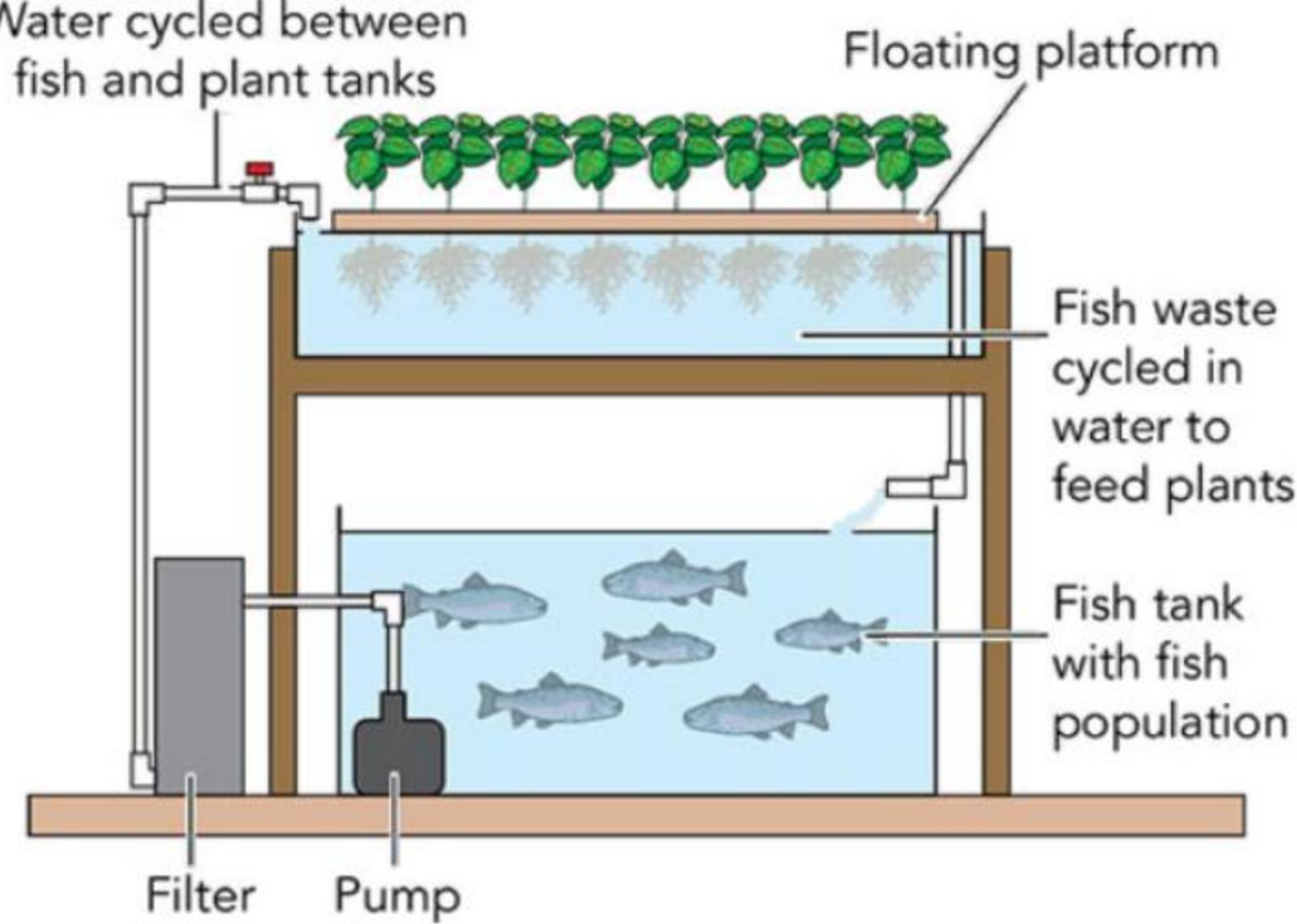
Like aeroponics, hydroponics and aquaponics are both soil-free farming practices. Plants are similarly fed using nutrient-rich water that is pumped around the system. Unlike aeroponics, the plant roots in both hydroponics and aquaponics are submerged in water.

Aquaponics goes one step further, with the system including fish. This has a number of benefits:

- 1 It allows the farmer to farm fish as a source of protein, in addition to the crops being grown.
- 2 The fish help to feed the plants. The fish process the nitrogen in the fish food, excreting it into the water. Bacteria then convert the excrement to nitrates. These nitrates are circulated in the water to the roots of the plants, which can absorb the nitrates and use them for growth.
- 3 The plants help to clean the water, which is circulated back to the fish, helping to maintain a healthy fish population.



Source C: A hydroponics system



Source **D**: An aquaponics system

In comparison to traditional arable farming, the benefits of aeroponics, hydroponics, and aquaponics are:

- it uses up to 98% less land
- it uses up to 95% less water
- it allows for year-round production; as the environment is controlled, food can be produced in all seasons
- it is more efficient
- it is safer for the consumer as no harmful chemicals are used.

While in comparison to traditional arable farming, the limitations of aeroponics, hydroponics, and aquaponics include:

- the high cost of setting up the systems
- the high maintenance of the systems
- the high level of technical knowledge required
- the rapid spread of diseases that harm the crops if bacteria gets into the system.

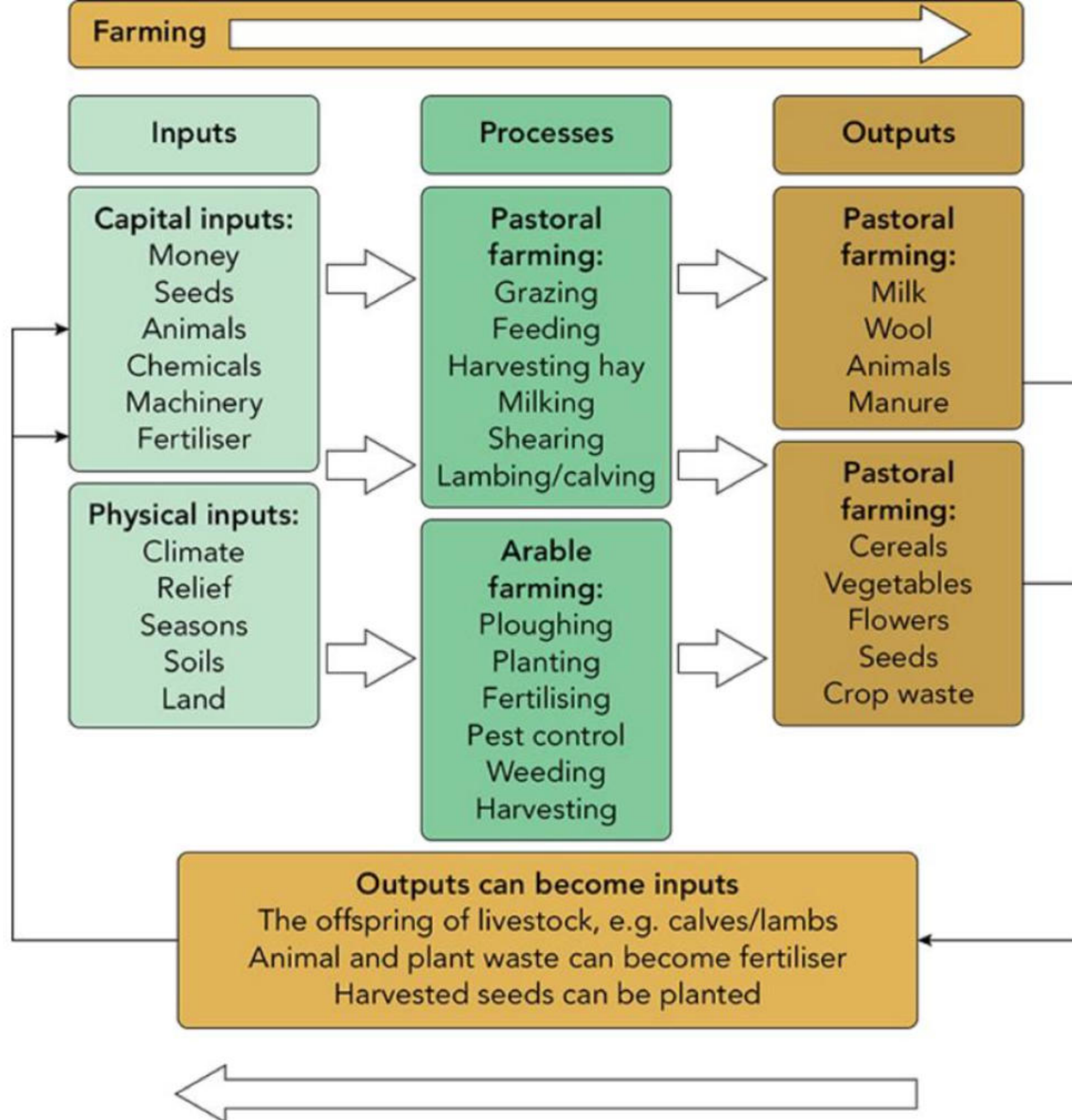
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TASK 2: Study Sources **B** **C** **D**

- What is the main benefit of aquaponics to a farmer, in comparison to aeroponics and hydroponics?
- Name two similarities and one difference between hydroponics and aeroponics.
- Discuss the limitations of these types of farming with a partner. Then, write 100-150 words explaining why a subsistence farmer may find it challenging to use a hydroponics, aquaponics or aeroponics system to grow crops.

Farming systems

All farming follows a system of **inputs**, **processes**, and **outputs**. Source E gives some examples of the types of inputs, processes, and outputs one can find in different types of farming.



Source **E**: A flow diagram showing the inputs, processes, and outputs of farming systems

► Image Description

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TASK 3: Study Source **E**

- Name two outputs a farmer could potentially get from farming chickens.
- Describe how the inputs and outputs for subsistence farming may differ from the inputs and outputs from commercial farming.
- Explain how an output from a farm can become an input to the same farm.
- Do you think that meat should be represented as a standard output from a pastoral farm? Consider the process required to produce meat when giving your answer.
- If you were to start a farm in your region, what do you think the main limitations would be? Consider the inputs, such as the space available, the climate, and the equipment you would need. What type of farming would you need to employ?

PROJECT

In small groups, design an indoor vegetable garden. You can research online for different ideas. Consider the following factors when you are designing the garden:

- the amount of space you have
- where you would put the garden (in a window, on a shelf with good light, on a rooftop, etc.)
- the materials available to you—you could challenge yourself to use only recycled materials
- the vegetables that you would grow
- the length of time it takes for those vegetables to grow from seed to harvesting
- the size of the plant when it is ready for harvesting.

Draw a diagram of your planned vegetable garden, explaining the step-by-step process that would be used for growing the plants. Include the inputs, processes, and outputs for your vegetable garden. You should write the process clearly, so that another person could follow it and grow the plants themselves.

As an extension activity, select the best vegetable garden idea. Build the garden and grow vegetables as a class.

REFLECTION

Did designing your own farming system help you to understand the relationship between inputs, processes, and outputs? Go back to Task 3e. Having done this project, can you now answer this question more easily?